

Insurance Customer & Revenue Analysis using SQL

Project Overview

Insurance companies rely heavily on data to understand customer behavior, optimize product offerings, and manage financial risk.

This project analyzes an insurance dataset using SQL to uncover insights related to:

- Customer distribution across plans
- Revenue contribution by product
- Customer segmentation
- High-value and high-risk policy identification

The objective is to simulate a real-world business scenario where data-driven decisions improve profitability and risk management.

Business Problem

The company wants to answer the following key questions:

- Which insurance plans drive the most revenue?
 - Are the most popular plans also the most profitable?
 - Which customer segments contribute the most value?
 - Are there potential high-risk policies based on coverage and duration?
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Dataset Description

Column	Description
age	Customer age
policy_term	Policy duration
ppt	Premium paying term
amount	Policy coverage amount

Column	Description
plan	Insurance plan type

Tools & Technologies

- PostgreSQL
 - pgAdmin
 - SQL (Aggregations, CASE, Window Functions)
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Key Analysis & Findings

Customer Distribution by Plan

- **Nonpar** has the highest number of customers (**67**)
- Followed by **ULIP (21)** and **Par (10)**
- **Health plan adoption is extremely low (2 customers)**

 Insight:

Customer preference is heavily skewed toward Nonpar plans.

Revenue Contribution by Plan

- **ULIP generates the highest revenue: 6,110,000**
- **Nonpar generates: 3,980,000**
- **Par contributes significantly less: 670,000**
- **Health plan revenue is negligible: 3,500**

 Insight:

The most popular plan (**Nonpar**) is NOT the most profitable.
ULIP dominates in revenue despite lower customer count.

Plan Performance Ranking

Plan Revenue Rank

ULIP 6,110,000 1

Nonpar 3,980,000 2

Par 670,000 3

Health 3,500 4

👉 Insight:

ULIP is a **high-value product** targeting fewer but higher-paying customers.

👤 High-Value Customers

- Top customers have coverage amounts of **300,000 – 500,000**
- Majority belong to **ULIP plans**

👉 Insight:

ULIP plans attract premium customers with higher policy values.

5 Risk Identification (High Coverage + Short Term)

- Policies with $\geq 500,000$ coverage and shorter durations identified
- These are concentrated in **ULIP plans**

👉 Insight:

High-value short-term policies represent **potential financial risk exposure**

🧠 Key Business Insights

- **Customer volume $\neq$ revenue generation**
 - Nonpar has the most customers but lower total revenue
- **ULIP is the most profitable product**
 - High-value customers drive revenue concentration
- **Revenue is highly concentrated in one plan (ULIP)**

- Indicates dependency risk
 - **Health plan underperformance**
 - Extremely low adoption and revenue
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Business Recommendations

1. Focus on High-Value Products

- Expand ULIP offerings
 - Target high-income customer segments
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2. Improve Low-Performing Plans

- Re-evaluate Health plan pricing and features
 - Increase marketing for underperforming products
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3. Risk Monitoring

- Closely monitor high-value short-term policies
 - Introduce stricter underwriting guidelines
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4. Balance Portfolio Strategy

- Reduce dependency on a single high-performing plan
 - Diversify revenue streams across products
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Future Enhancements

- Build interactive dashboards using Power BI
- Perform predictive modeling (customer segmentation, risk scoring)
- Integrate claims data for deeper risk analysis

- Conduct customer lifetime value (CLV) analysis
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Project Structure

insurance-sql-analysis/

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├── data/

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└─ Insurance.csv

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├── sql/

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└─ analysis_queries.sql

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├── outputs/

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└─ query_results_screenshots.png

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└─ README.md

Conclusion

This project demonstrates how SQL can be used to extract actionable business insights from structured data.

By analyzing customer behavior and revenue distribution, the project highlights the importance of:

- Data-driven decision making
- Product optimization
- Risk management

These insights can help insurance companies improve profitability while managing exposure effectively.